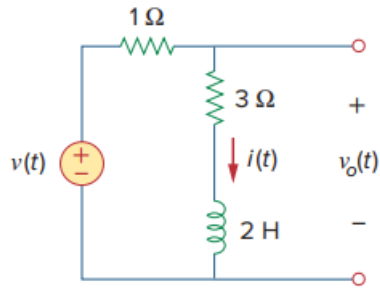


Problem

Consider the circuit of Fig. 7.97. Find $v_o(t)$ if $i(0) = 15$ A and $v(t) = 0$.



Step-by-step solution

Step 1 of 5

Consider that the circuit is source free network. Therefore, current flowing through the inductor is

$$i(t) = i(0)e^{-\frac{t}{\tau}} \dots\dots (1)$$

Calculate the time constant.

$$\tau = \frac{L}{R_{eq}} \dots\dots (2)$$

Step 2 of 5

Determine the equivalent resistance.

Consider the following circuit:

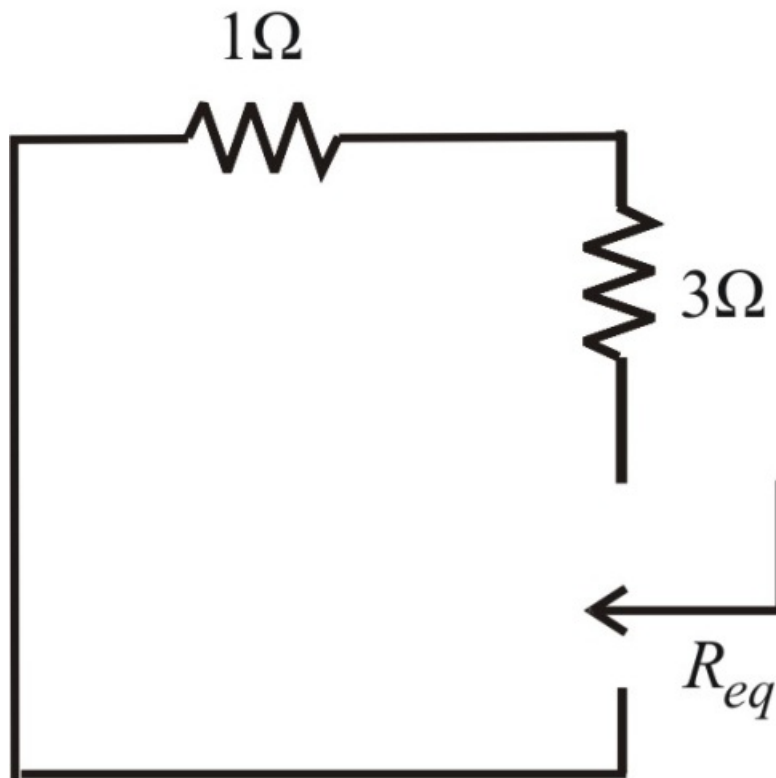


Figure 1

Step 3 of 5

Calculate the equivalent resistance when seen from the inductor as shown in figure 1.

$$R_{eq} = 3 + 1$$

$$R_{eq} = 4\ \Omega$$

Recall equation (2).

$$\tau = \frac{L}{R_{eq}}$$

Substitute 2 for L , and 4 for R_{eq} .

$$\tau = \frac{2}{4}$$

$$\tau = \frac{1}{2} \text{ sec}$$

Step 4 of 5

Recall equation (1).

$$i(t) = i(0)e^{-\frac{t}{\tau}}$$

Substitute 15 for $i(0)$, and $\frac{1}{2}$ for τ .

$$i(t) = 15e^{-2t} \text{ A}$$

Therefore, the current through inductor is

$$i(t) = 15e^{-2t} \text{ A}$$

Step 5 of 5

Calculate the output voltage.

$$\begin{aligned} v_o(t) &= 3i(t) + v_L(t) \\ &= 3i(t) + 2\frac{di(t)}{dt} \\ &= 3(15e^{-2t}) + 2\frac{d}{dt}(15e^{-2t}) \\ &= 45e^{-2t} + 2(15)(-2)e^{-2t} \\ &= (45 - 60)e^{-2t} \\ v_o(t) &= -15e^{-2t} \text{ V} \end{aligned}$$

Therefore, the output voltage for the given circuit is, $v_o(t) = -15e^{-2t} \text{ V}$.